

MATH-2111

Tutorial 3: Linear Transformations, Kernel, Range & Independence

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Problem 1: Images from Basis Vectors

Problem 1

Let

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{y}_1 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \quad \mathbf{y}_2 = \begin{bmatrix} -1 \\ 6 \end{bmatrix}.$$

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation satisfying

$$T(\mathbf{e}_1) = \mathbf{y}_1, \quad T(\mathbf{e}_2) = \mathbf{y}_2.$$

Standard matrix?

- (a) Find $T\left(\begin{bmatrix} 5 \\ -3 \end{bmatrix}\right)$.
- (b) Find $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right)$.

Solution to Problem 1

It is efficient to establish the general formula in part (b) first. Every vector in $\mathbb{R}^2$ has the standard-basis expansion

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2.$$

By linearity,

$$\begin{aligned} T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) &= x_1 T(\mathbf{e}_1) + x_2 T(\mathbf{e}_2) \\ &= x_1 \begin{bmatrix} 2 \\ 5 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 6 \end{bmatrix} = \boxed{\begin{bmatrix} 2x_1 - x_2 \\ 5x_1 + 6x_2 \end{bmatrix}}. \end{aligned}$$

Hence the standard matrix is

$$\boxed{A = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2)] = \begin{bmatrix} 2 & -1 \\ 5 & 6 \end{bmatrix}}.$$

In particular,

$$T\left(\begin{bmatrix} 5 \\ -3 \end{bmatrix}\right) = 5 \begin{bmatrix} 2 \\ 5 \end{bmatrix} - 3 \begin{bmatrix} -1 \\ 6 \end{bmatrix} = \boxed{\begin{bmatrix} 13 \\ 7 \end{bmatrix}}.$$

Problem 2: Finding All Vectors Mapped to Zero

Problem 2

Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ be the matrix transformation $T(\mathbf{x}) = A\mathbf{x}$, where

$$A = \begin{bmatrix} 1 & 3 & 9 & 2 \\ 1 & 0 & 3 & -4 \\ 0 & 1 & 2 & 3 \\ -2 & 3 & 0 & 5 \end{bmatrix}.$$

Find all $\mathbf{x} \in \mathbb{R}^4$ such that $T(\mathbf{x}) = \mathbf{0}$. Equivalently, determine $\text{Ker}(T) = \text{Nul}(A)$ by solving $A\mathbf{x} = \mathbf{0}$.

$$\text{Nul}(A) = \{\mathbf{x} \in \mathbb{R}^4 \mid A\mathbf{x} = \mathbf{0}\}$$

$\Downarrow$ $\text{ker}(T)$: kernel, map to "0" at target space

Solution to Problem 2: Row Reduction

Start with the coefficient matrix. Apply $R_2 \leftarrow R_2 - R_1$, $R_4 \leftarrow R_4 + 2R_1$, and then interchange R_2 and R_3 :

$$\begin{bmatrix} 1 & 3 & 9 & 2 \\ 1 & 0 & 3 & -4 \\ 0 & 1 & 2 & 3 \\ -2 & 3 & 0 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 9 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & -3 & -6 & -6 \\ 0 & 9 & 18 & 9 \end{bmatrix}.$$

Next use $R_3 \leftarrow R_3 + 3R_2$ and $R_4 \leftarrow R_4 - 9R_2$:

$$\sim \begin{bmatrix} 1 & 3 & 9 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & -18 \end{bmatrix}.$$

After scaling the third row, eliminating the fourth row, and clearing the pivot columns,

$$\text{rref}(A) = \begin{bmatrix} 1 & 0 & 3 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Solution to Problem 2: Parametric Form

The reduced equations are

$$x_1 + 3x_3 = 0, \quad x_2 + 2x_3 = 0, \quad x_4 = 0.$$

Let $x_3 = t$. Then

$$\mathbf{x} = \begin{bmatrix} -3t \\ -2t \\ t \\ 0 \end{bmatrix} = t \begin{bmatrix} -3 \\ -2 \\ 1 \\ 0 \end{bmatrix}.$$

Therefore

$$\text{Ker}(T) = \text{Nul}(A) = \left\{ t \begin{bmatrix} -3 \\ -2 \\ 1 \\ 0 \end{bmatrix} : t \in \mathbb{R} \right\}.$$

Direct substitution verifies that $A(-3, -2, 1, 0)^T = \mathbf{0}$. Since x_3 is the only free variable, this one direction generates the entire kernel.

Problem 3: Testing Membership in the Range

Problem 3

Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be the matrix transformation $T(\mathbf{x}) = A\mathbf{x}$, where

$$A = \begin{bmatrix} 1 & -4 & 7 & -5 \\ 0 & 1 & -4 & 3 \\ 2 & -6 & 6 & -4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

Is $\mathbf{b} \in \text{Range}(T) = \text{Col}(A)$? Explain why or why not.



Column Space

Solution to Problem 3

Membership in the range is equivalent to consistency of $A\mathbf{x} = \mathbf{b}$. Row reduce the augmented matrix:

$$\left[\begin{array}{cccc|c} 1 & -4 & 7 & -5 & -1 \\ 0 & 1 & -4 & 3 & 1 \\ 2 & -6 & 6 & -4 & 0 \end{array} \right] \xrightarrow{R_3 \leftarrow R_3 - 2R_1} \left[\begin{array}{cccc|c} 1 & -4 & 7 & -5 & -1 \\ 0 & 1 & -4 & 3 & 1 \\ 0 & 2 & -8 & 6 & 2 \end{array} \right].$$

Then $R_3 \leftarrow R_3 - 2R_2$ and $R_1 \leftarrow R_1 + 4R_2$ give

$$\left[\begin{array}{cccc|c} 1 & 0 & -9 & 7 & 3 \\ 0 & 1 & -4 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

There is no contradictory row, so the system is consistent. For example, choosing $x_3 = x_4 = 0$ gives

$$\mathbf{x} = \begin{bmatrix} 3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad A\mathbf{x} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \mathbf{b}.$$

Hence

$$\mathbf{b} \in \text{Range}(T) = \text{Col}(A).$$

Problem 4: Uniqueness Implies Independence

Problem 4

Suppose A is an $m \times n$ matrix with the property that, for every $\mathbf{b} \in \mathbb{R}^m$, the equation

$$A\mathbf{x} = \mathbf{b}$$

has at most one solution.

Use the definition of linear independence to explain why the columns of A must be linearly independent.

Solution to Problem 4

Write the columns of A as

$$A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n] .$$

To test linear independence, suppose

$$c_1 \mathbf{a}_1 + c_2 \mathbf{a}_2 + \cdots + c_n \mathbf{a}_n = \mathbf{0} .$$

With $\mathbf{c} = (c_1, \dots, c_n)^T$, this relation is exactly

$$A\mathbf{c} = \mathbf{0} .$$

For $\mathbf{b} = \mathbf{0}$, the vector $\mathbf{x} = \mathbf{0}$ is always a solution. The hypothesis says that the equation has at most one solution, so it cannot have any other solution. Consequently,

$$\mathbf{c} = \mathbf{0}, \quad \text{that is,} \quad c_1 = \cdots = c_n = 0 .$$

By the definition of linear independence,

$\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ is linearly independent.

Logical point. The full hypothesis is stronger than needed: only uniqueness for $A\mathbf{x} = \mathbf{0}$ is required.

Problem 5: Vanishing on a Spanning Set

Problem 5

Suppose vectors $\mathbf{v}_1, \dots, \mathbf{v}_p$ span $\mathbb{R}^n$, and let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation. Suppose

$$T(\mathbf{v}_i) = \mathbf{0} \quad \text{for } i = 1, \dots, p.$$

Show that T is the zero transformation. That is, prove that

$$T(\mathbf{x}) = \mathbf{0} \quad \text{for every } \mathbf{x} \in \mathbb{R}^n.$$

Solution to Problem 5

Let $\mathbf{x} \in \mathbb{R}^n$ be arbitrary. Since $\mathbf{v}_1, \dots, \mathbf{v}_p$ span $\mathbb{R}^n$, there exist scalars $c_1, \dots, c_p$ such that

$$\mathbf{x} = c_1 \mathbf{v}_1 + \dots + c_p \mathbf{v}_p.$$

Using linearity,

$$\begin{aligned} T(\mathbf{x}) &= T(c_1 \mathbf{v}_1 + \dots + c_p \mathbf{v}_p) \\ &= c_1 T(\mathbf{v}_1) + \dots + c_p T(\mathbf{v}_p) \\ &= c_1 \mathbf{0} + \dots + c_p \mathbf{0} \\ &= \mathbf{0}. \end{aligned}$$

Because $\mathbf{x}$ was arbitrary,

$T(\mathbf{x}) = \mathbf{0} \text{ for every } \mathbf{x} \in \mathbb{R}^n.$

Therefore T is the zero transformation.

Why a spanning set is enough

The vectors need not be linearly independent. Once their span is the entire domain, linearity forces the values of T everywhere from its values on those vectors.

Problem 6: Consistency, Matrix, and Preimages

Problem 6

Let

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{u}_4 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

A linear map $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is claimed to satisfy

$$T(\mathbf{u}_1) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad T(\mathbf{u}_2) = \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \quad T(\mathbf{u}_3) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad T(\mathbf{u}_4) = \begin{bmatrix} k \\ 4 \end{bmatrix}.$$

- (a) Determine the value of k that is necessary and sufficient for all four claims to be consistent with linearity.
- (b) For that value of k , find the standard matrix of T .
- (c) Decide whether T is onto and whether it is one-to-one.
- (d) Find all $\mathbf{x}$ satisfying $T(\mathbf{x}) = T(\mathbf{u}_1)$.

Solution to Problem 6(a): Consistency

The input vectors satisfy the relation

$$\mathbf{u}_4 = \mathbf{u}_1 + \mathbf{u}_2 - 2\mathbf{u}_3.$$

Every linear relation among the inputs must be preserved by a linear map. Therefore

$$\begin{aligned} T(\mathbf{u}_4) &= T(\mathbf{u}_1) + T(\mathbf{u}_2) - 2T(\mathbf{u}_3) \\ &= \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}. \end{aligned}$$

Thus the required value is

$$\boxed{k = -1}.$$

This condition is also sufficient. Indeed,

$$\mathbf{e}_3 = \mathbf{u}_3, \quad \mathbf{e}_1 = \mathbf{u}_1 - \mathbf{u}_3, \quad \mathbf{e}_2 = \mathbf{u}_2 - \mathbf{u}_3,$$

so $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ form a basis of $\mathbb{R}^3$ and determine a unique linear map. The value $k = -1$ makes the fourth prescribed image agree with that map.

Solution to Problem 6(b): Standard Matrix

From the standard-basis identities,

$$T(\mathbf{e}_1) = T(\mathbf{u}_1) - T(\mathbf{u}_3) = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

$$T(\mathbf{e}_2) = T(\mathbf{u}_2) - T(\mathbf{u}_3) = \begin{bmatrix} -2 \\ 3 \end{bmatrix}, \quad T(\mathbf{e}_3) = T(\mathbf{u}_3) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Hence the standard matrix is

$$A = \begin{bmatrix} 1 & -2 & 1 \\ 1 & 3 & 0 \end{bmatrix}.$$

Internal check

For $k = -1$, direct multiplication gives $A\mathbf{u}_i = T(\mathbf{u}_i)$ for all four prescribed vectors.

Solution to Problem 6(c): Onto and One-to-One

Row reduction gives

$$\begin{bmatrix} 1 & -2 & 1 \\ 1 & 3 & 0 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{bmatrix} 1 & -2 & 1 \\ 0 & 5 & -1 \end{bmatrix}.$$

There is a pivot in every row, so the columns span $\mathbb{R}^2$ and

$$\boxed{T \text{ is onto } \mathbb{R}^2.}$$

There is not a pivot in every column, so the homogeneous equation has a free variable. From the echelon system,

$$5x_2 - x_3 = 0, \quad x_1 - 2x_2 + x_3 = 0.$$

Setting $x_2 = t$ gives $x_3 = 5t$ and $x_1 = -3t$. Hence

$$\text{Ker}(T) = \left\{ t \begin{bmatrix} -3 \\ 1 \\ 5 \end{bmatrix} : t \in \mathbb{R} \right\}.$$

Since the kernel contains nonzero vectors,

$$\boxed{T \text{ is not one-to-one.}}$$

Solution to Problem 6(d): All Preimages

Because T is linear,

$$T(\mathbf{x}) = T(\mathbf{u}_1) \iff T(\mathbf{x} - \mathbf{u}_1) = \mathbf{0} \iff \mathbf{x} - \mathbf{u}_1 \in \text{Ker}(T).$$

Using

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \text{Ker}(T) = \text{Span} \left\{ \begin{bmatrix} -3 \\ 1 \\ 5 \end{bmatrix} \right\},$$

we obtain

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} -3 \\ 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 - 3t \\ t \\ 1 + 5t \end{bmatrix}, \quad t \in \mathbb{R}.$$

Geometrically, the full preimage is the affine line through $\mathbf{u}_1$ parallel to $\text{Ker}(T)$.